Post-Processing Load Estimation Methods for Simplified Urban Building Energy Modelling

Archetype-based Urban Building Energy Modelling (UBEM) frequently relies on simplified thermodynamic representations, such as the EnergyPlus "Ideal Loads" system, to accelerate simulation runtime across thousands of individual building geometries. While this approach effectively computes the thermal energy required to maintain indoor temperature setpoints, it inherently bypasses the modeling of physical secondary and tertiary systems, such as air-handling units, fluid distribution networks, service water heaters, and process loads. Consequently, critical auxiliary end-uses—including supply and extract fans, hydronic pumps, domestic hot water (DHW), and commercial refrigeration—are omitted. Because these neglected end-uses typically represent more than 40% of the total energy performance gap in single-building validation exercises, establishing defensible, simplified post-processing algorithms is essential to restore these loads without compromising the computational speed of the UBEM. This report presents validated, citation-backed equations, empirical coefficients, and engineering models designed to reconstruct these loads as a deterministic post-processing layer.

Service Water Heating and Domestic Hot Water Loads

The thermal demand associated with heating domestic water is decoupled from the building's space-conditioning thermodynamic balance, making it highly suitable for deterministic post-processing. The primary challenge lies in converting daily volume-draw expectations into hourly energy vectors. The energy required to raise the temperature of incoming cold water mains to the municipal or archetype-specific setpoint is calculated using a thermodynamic energy balance:

$$E_{\mathrm{SWH}} = \frac{V_{\mathrm{draw}} \cdot \rho_{\mathrm{w}} \cdot C_{p,\mathrm{w}} \cdot (T_{\mathrm{set}} - T_{\mathrm{in}})}{3,600 \cdot \eta_{\mathrm{sys}}}$$

In this formulation, E_{SWH} represents the daily or hourly water heating energy consumption in kilowatthours (kWh). The volume of hot water consumed, V_{draw}, is expressed in litres (L), while ρ_{w} represents the density of water ($\approx 1.0\ \mathrm{kg/L}$ at standard temperature and pressure) and $C_{p,\mathrm{w}}$ is the specific heat capacity ($\approx 4.184\ \mathrm{kJ/kg \cdot K}$)[1]. The water heater setpoint, T_{set}, and the incoming mains water temperature, T_{in}, are both defined in degrees Celsius ($^{\circ}\mathrm{C}$). The system efficiency, η_{sys}, is a decimal fraction representing the combined product of the water heater's thermal efficiency (η_{thermal}) and a distribution system multiplier (η_{dist}) designed to account for piping conduction losses and recirculation loop degradation[2]. For imperial configurations, the thermal load is calculated in British thermal units (Btu/day) using the following equation:

$$\text{DHWL} = V_{\text{draw}} \cdot \gamma_{\text{w}} \cdot (T_{\text{set}} - T_{\text{in}})$$

where V_{draw} represents the average gallons per day (AGPD), γ_{w} represents the nominal weight density of water ($\approx 8.28\ \text{lb/gal}$ to $8.33\ \text{lb/gal}$ depending on temperature), and temperatures are expressed in degrees Fahrenheit ($^\circ\text{F}$)[1]. Under ASHRAE Standard 90.2-2001, the domestic hot water load for multi-family living units is estimated dynamically based on the number of units and bedrooms[4]:

$$\text{DHWL} = ((30 \times \text{Units}) + (10 \times \text{Bedrooms})) \cdot \text{AGPD} \cdot 8.28 \cdot (135 - T_{\text{in}})$$

The system's real-world thermal efficiency is heavily degraded by distribution losses. Empirical studies indicate that point-of-use systems reduce distribution losses significantly, operating with a baseline loss multiplier of 0.82, whereas uninsulated central recirculation systems exhibit a high-loss multiplier of 1.52 due to continuous thermal dissipation through uninsulated pipe walls[2]. Incorporating pipe insulation brings the multiplier down to 0.92, while incorporating time-clock or temperature-based recirculation controls achieves multipliers of 1.28 and 1.05 respectively[2]. Furthermore, ASHRAE Chapter 50 dictates that public lavatories restrict hot water delivery to a maximum of 43°C (110°F) to prevent scalding and save energy, while commercial food services require setpoints of 60°C (140°F) at the primary tank, often coupled with local booster water heaters designed to deliver 82°C to 91°C (180°F to 195°F) sanitizing water[5].

Table 1: Service Water Heating (SWH) Demand Coefficients and Sizing Parameters

Archetype	Sizing Basis	Daily Coefficie	Daily Coefficie	Standard Setpoint	Standard Draw	Key Referenc

Building Type		nt (Metric)	nt (Imperial)	(Tset)	Schedule Profile	e
Large Office	Floor Area	$57.04\ \mathrm{L}/($	$1.40\ \mathrm{gal}/($	$49°\mathrm{C}$ ($120°\mathrm{F}$)	Daytime occupancy peak (08:00–17:00), zero weekend use	LBL-37398[8]
Small Office	Floor Area	$57.04\ \mathrm{L}/($	$1.40\ \mathrm{gal}/($	$49°\mathrm{C}$ ($120°\mathrm{F}$)	Daytime occupancy peak (08:00–17:00), zero weekend use	LBL-37398[8]
Primary School	Occupancy	$2.27\ \mathrm{L}/(\mathrm{s}$	$0.60\ \mathrm{gal}/($	$43°\mathrm{C}$ ($110°\mathrm{F}$)	Diurnal school hours peak, closed on weekends	ASHRAE Chapter 50[8]
Secondary School	Occupancy	$6.81\ \mathrm{L}/(\mathrm{s}$	$1.80\ \mathrm{gal}/($	$43°\mathrm{C}$ ($110°\mathrm{F}$)	Diurnal school hours, gym-midday shower spikes	ASHRAE Chapter 50[8]
Stand-alone	Floor Area	$59.07\ \mathrm{L}/($	$1.45\ \mathrm{gal}/($	$49°\mathrm{C}$ (	Smooth schedule matching	LBL-37398[8]

Retail				$120^{\circ}\mathrm{F}$)	store operating hours	
Supermarket / Grocery	Floor Area	$59.07\ \mathrm{L}/($	$1.45\ \mathrm{gal}/($	$60^{\circ}\mathrm{C}$ ($140^{\circ}\mathrm{F}$)	Constant low draw, high late-night cleaning peaks	LBL-37398[8]
Full-Service Restaurant	Occupancy	$9.08\ \mathrm{L}/(\mathrm{n}$	$2.40\ \mathrm{gal}/($	$60^{\circ}\mathrm{C}$ ($140^{\circ}\mathrm{F}$)	Heavy lunch (12:00) and dinner (18:00–21:00) peaks	ASHRAE Chapter 50[6]
Quick-Service Rest.	Occupancy	$2.65\ \mathrm{L}/(\mathrm{n}$	$0.70\ \mathrm{gal}/($	$60^{\circ}\mathrm{C}$ ($140^{\circ}\mathrm{F}$)	Extended operational hours, continuous draws	ASHRAE Chapter 50[8]
Large Hotel / Lodging	Occupancy	$79.85\ \mathrm{L}/($	$1.96\ \mathrm{gal}/($	$60^{\circ}\mathrm{C}$ ($140^{\circ}\mathrm{F}$)	High bimodal peaks (07:00–09:00 and 18:00–21:00)	LBL-37398[8]
Hospital	Occupancy	$69.65\ \mathrm{L}/($	$18.40\ \mathrm{gal},$	$49^{\circ}\mathrm{C}$ ($120^{\circ}\mathrm{F}$)	Relatively flat 24-hour baseline, small morning	ASHRAE Chapter 50[8]

					peak	
Warehouse	Floor Area	$37.89\ \mathrm{L}/($	$0.93\ \mathrm{gal}/($	$49^{\circ}\mathrm{C}$ ($120^{\circ}\mathrm{F}$)	Low draw, clustered around operational shift changes	CADMUS [8]
Multi-Family Mid-Rise	Floor Area	$99.81\ \mathrm{L}/($	$2.45\ \mathrm{gal}/($	$49^{\circ}\mathrm{C}$ ($120^{\circ}\mathrm{F}$)	Bimodal residential draw (morning and evening)	LBL-37398 [8]

Fan and Pump Energy Auxiliary Load Quantification

Auxiliary HVAC loads—comprising air-handling fans and hydronic loops—can be mapped from simulated hourly ideal heating and cooling sensible thermal loads[11]. This translation leverages physics-based engineering definitions to convert thermal power demands into equivalent electrical fan and pump energy vectors[12].

Airside Fan Power Formulation

Specific Fan Power (SFP) acts as a standardized index to quantify the energy performance of ventilation loops[12]. SFP represents the ratio of combined fan electrical power consumption to the gross air volumetric flow rate delivered[12]:

$$\mathrm{SFP} = \frac{\sum P_{\mathrm{fan}}}{Q_{\mathrm{air}}} = \frac{\Delta p_{\mathrm{system}}}{\eta_{\mathrm{fan}}}$$

where SFP is expressed in watts per litre per second ($\mathrm{W}/(\mathrm{L}/\mathrm{s})$) or kilowatts per cubic metre per second ($\mathrm{kW}/(\mathrm{m}^3/\mathrm{s})$), which are equivalent units[12]. The total electrical power input to the fan, P_{fan} (W), includes motor, belt drive, and variable speed drive (VSD) losses[12]. The volumetric airflow rate, Q_{air} (L/s), is determined from the zone's sensible thermal cooling or heating load (q_{sensible}) under design air loop temperature differences[11]:

$$Q_{\text{air}} = \frac{q_{\text{sensible}}}{\rho_{\text{air}} \cdot C_{p,\text{air}} \cdot (T_{\text{zone}} - T_{\text{supply}})}$$

Using standard air density ($\rho_{\text{air}} \approx 1.2\ \text{kg/m}^3$) and specific heat capacity ($C_{p,\text{air}} \approx 1.006\ \text{kJ/kg} \cdot \text{K}$), the volumetric specific product $\rho_{\text{air}} \cdot C_{p,\text{air}}$ yields approximately $1.2\ \text{W/(L/s} \cdot \text{K)}$ [11]. A standard temperature differential ($T_{\text{zone}} - T_{\text{supply}}$) of $11.1\ \text{K}$ (20°F) is assumed for cooling, and $16.7\ \text{K}$ (30°F) is assumed for heating. For variable air volume (VAV) systems, the fan power scales non-linearly under part-load conditions. The actual electrical power used during any hour is the design fan power adjusted by a part-load performance fraction (f_{PL}) as specified in ASHRAE Standard 90.1 Appendix G[16]:

$$f_{\text{PL}} = 0.0013 + 0.1470 \cdot \chi + 0.2629 \cdot \chi^2 + 0.5894 \cdot \chi^3$$

where χ represents the airflow fraction ($Q_{\text{hourly}} / Q_{\text{design}}$)[16]. Under frequency inverter (VSD) control, empirical auditing reveals that the fan power is proportional to the average airflow rate raised to an exponent of 1.5 to 2.0 depending on static pressure control resets, rather than a perfect cubic relation[18].

Hydronic Pump Power Formulation

Specific Pump Power (SPP) normalizes the energy required to circulate heating and cooling fluids through distribution pipe networks[13]:

$$\text{SPP} = \frac{\sum P_{\text{pump}}}{Q_{\text{fluid}}} = \frac{\rho_{\text{fluid}} \cdot g \cdot H}{\eta_{\text{pump}} \cdot \eta_{\text{motor}}}$$

where SPP is expressed in kilowatts per litre per second (kW/(L/s)) or equivalently in watts per litre per second (W/(L/s))[13]. The volumetric flow rate of water, Q_{fluid} (L/s), is determined from the dynamic thermal heating or cooling load requirements at the zone coil (q_{thermal})[20]:

$$Q_{\text{fluid}} = \frac{q_{\text{thermal}}}{\rho_{\text{fluid}} \cdot C_{p,\text{fluid}} \cdot \Delta T_{\text{fluid}}}$$

Standard water heat capacity limits the volumetric specific product $\rho_{\text{fluid}} \cdot C_{p,\text{fluid}}$ to

approximately $4,184\ \mathrm{W/(L/s \cdot K)}$ [19]. The standard hydronic loop temperature differential (ΔT_{fluid}) is assumed to be $5.55\ \mathrm{K}$ ($10^{\circ}\mathrm{F}$) for chilled water systems and $11.1\ \mathrm{K}$ ($20^{\circ}\mathrm{F}$) for hot water networks. Under part-load scenarios, secondary variable-flow pump power scales in direct proportion to the system flow rate raised to an exponent of 2.0 to 2.5 depending on dynamic differential pressure control[19].
Moreover, the inter-dependencies of internal loads heavily influence these auxiliary systems. Empirical modeling shows that reducing internal heat gains—for instance, by decreasing lighting power densities (LPD)—results in a 2.88% reduction in space cooling loads, a 0.26% decrease in HVAC fan power, and a disproportionate 13.54% reduction in chilled-water pump energy[21]. This non-linear pump sensitivity occurs because secondary chilled-water loops react directly and proportionally to reduced heat dissipation, whereas ventilation fans are bound by statutory minimum outdoor air flow requirements and must maintain their operating speeds regardless of load[21].

Table 2: Fan and Pump Energy Sizing and Scaling Methods

Method / Auxiliary Parameter	Coefficient / Sizing Rule	Applicable HVAC System	Key Reference
SFP - Central Mech. Ventilation	$2.50\ \mathrm{W/(L/s}$	Central system with heating, cooling, heat recovery	EN 13779 / ISO 5801[16]
SFP - Central Mech. Ventilation	$2.00\ \mathrm{W/(L/s}$	Central system with heating and cooling only	EN 13779 / DesignBuilder[16]
SFP - Central CAV / VAV Baseline	$1.80\ \mathrm{W/(L/s}$	Standard ducted multi-zone systems	ASHRAE 90.1 / Tas[16]
SFP - Fan Coil Installation	$0.80\ \mathrm{W/(L/s}$	Weighted multi-unit average fan coil systems	BS 8850 / ClimateStudio[16]
SFP - High-Efficiency	$0.40\ \mathrm{W/(L/s}$	Advanced low-pressure-drop	UK Approved Document L[24]

Fan Coil		installations	
SFP - Local Decentralised Vent.	$0.50\ \mathrm{W/(L/s}$	In-space window, wall, or local roof fans	ISO 5801[16]
SFP - Remote Decentralised Vent.	$1.20\ \mathrm{W/(L/s}$	Ceiling-void or inline roof exhaust fans	ISO 5801[16]
SPP - Chilled Water Circulation	$300\ \mathrm{W/(L/s}$	Central chilled water distribution loops	ASHRAE 90.1 / SPP Guide[13]
SPP - Hot Water Hydronic Loop	$301\ \mathrm{W/(L/s}$	Central gas/oil boiler heating loops	ASHRAE 90.1 / SPP Guide[13]
SPP - Low-Head Radiant Loop	$142.6\ \mathrm{W/(L/s}$ ($9\ \mathrm{W/gpm}$)	Radiant slab concrete-core heating/cooling	Gayeski / MIT LLCS[25]
SFP Part-Load Power Fraction	$f_{\mathrm{PL}} = 0.0013 + 0$	VAV systems with VSD motor controls	ASHRAE 90.1 Appendix G[16]
SPP Part-Load Pumping Fraction	$f_{\mathrm{PL}} =$	Variable-flow pumping hydronic systems	Chalmers University[19]

Commercial Refrigeration Energy Profiles

Commercial refrigeration stands out as a highly intensive, process-driven end-use with a continuous base-load profile that runs independently of occupancy schedules[26]. While it is fundamentally driven by internal retail activities, refrigeration remains dynamically linked to ambient indoor humidity and dry-bulb conditions because moist zone air increases frost accumulation on low-temperature evaporator coils[26].

Table 3: Commercial Refrigeration Electricity Intensities by Archetype

Archetype	Metric EUI	Imperial EUI	Baseline Appliance and	Key

Building Type	(kWh/(m2·yr))	(kBtu/(ft2·yr))	System Configurations	Reference
Supermarket / Grocery	$331.0 -$	$105.0 -$	26 display cases (LT/MT), 10 walk-in freezers, centralized compressor racks	AEDG / CBECS[26]
Convenience Store	684.6	216.3	Walk-in self-contained beverage coolers, ice machines, high unit-to-area density	PNNL OpenStudio Prototype[29]
Food Service / Restaurant	128.0	40.6	Walk-in coolers (78%), commercial ice makers (74%), display cabinets (66%)	CBECS 2018 Food Service[6]
Refrigerated Warehouse	$78.8 -$	$25.0 -$	Industrial blast freezers, continuous cold storage, evaporator fan arrays	CBECS 2018 Warehouse[30]
Large Office	$1.6 -$	$0.5 -$	Compact breakroom refrigerators (73%),	CBECS 2018 Office[31]

			drinking water coolers	
Hospital / Healthcare	$3.2 -$	$1.0 -$	Laboratory refrigeration, blood storage freezers, main kitchen storage	CBECS 2018 Health[32]
Vacant Building	1.6	0.5	Minimum compact security refrigeration, zero commercial refrigeration	CBECS 2018 Vacant[34]

Baseline End-Use EUI Splits by Archetype

To allow the back-calculation of auxiliary and service water heating energy vectors from basic UBEM spatial databases, a comprehensive breakdown of end-use fuel fractions has been synthesised. These ratios are derived from the latest 2018 CBECS datasets and the PNNL Commercial Prototype Building models[30].

Table 4: Consolidated End-Use Energy Fractions by Archetype (%)

Building Archetype	**Space Heat**	**Space Cool**	**Vent. (Fans)**	**Pumps**	**SWH (DHW)**	**Lighting**	**Equip. (Plug)**	**Refrig.**	**Cooking / Other**
Large Office [cite: 35, 36]	30.0%	14.0%	11.0%	3.5%	1.5%	12.0%	27.0%	0.5%	0.5%

Small Office [cite: 35, 37]	35.0%	10.0%	12.0%	1.0%	1.0%	15.0%	25.0%	0.5%	0.5%
Primary School [cite: 35, 38]	38.0%	8.0%	12.0%	3.0%	3.0%	14.0%	18.0%	1.0%	3.0%
Secondary School [cite: 35, 38]	36.0%	10.0%	11.0%	3.0%	4.0%	14.0%	19.0%	1.0%	2.0%
Stand-alone Retail [cite: 35, 37]	28.0%	13.0%	12.0%	1.5%	1.5%	22.0%	18.0%	1.5%	2.5%
Supermarket [cite: 26]	9.0%	6.0%	10.0%	1.0%	1.0%	13.0%	8.0%	50.0%	2.0%
Full-Service Rest.	12.0%	7.0%	7.0%	1.5%	8.0%	5.0%	9.0%	15.0%	35.5%

[cite: 6, 35]									
Large Hotel [cite: 9, 35]	25.0%	14.0%	8.0%	3.0%	20.0%	10.0%	15.0%	2.0%	3.0%
Hospital [cite: 8, 35]	40.0%	14.0%	10.0%	3.5%	3.0%	11.0%	16.0%	1.0%	1.5%
Warehouse [cite: 30, 38]	51.0%	3.0%	9.0%	1.0%	1.0%	10.0%	4.0%	5.0%	16.0%
Mid-Rise Apt. [cite: 4, 35]	28.0%	11.0%	5.0%	1.0%	23.0%	8.0%	22.0%	1.0%	1.0%

Computational Feasibility of Post-Processing Integration

Implementing service and auxiliary loads as a deterministic post-processing layer represents a highly practical strategy for large-scale UBEM, though its applicability varies depending on the physical coupling between end-uses and the building's thermal envelope.

First, service water heating (SWH) can be post-processed with high confidence. Because hot water consumption is a process load driven primarily by occupant behavior, it can be calculated hourly based on archetype-specific occupancy schedules and outdoor dry-bulb temperatures, which dictate the seasonal variation in mains water temperatures. However, this decoupled approach fails to capture the local standby losses of the water heater and its hot water distribution piping. In reality, these distribution heat losses act as uncontrolled, continuous internal sensible heat gains within the thermal zones. This omission slightly inflates the simulated space-heating loads and underpredicts the cooling loads, especially in buildings with central recirculation systems where distribution losses are continuous.

Second, ventilation fan energy and hydronic circulation pump energy are moderately suited for decoupled post-processing. Because the volumetric flow rates of air and water are calculated directly from the simulated sensible loads at each timestep, the SFP and SPP multipliers can be applied deterministically to compute auxiliary electricity consumption. The main limitation is that this post-processing approach does not account for fan heat pick-up. In physical HVAC air loops, a portion of the electrical energy consumed by the fan motor is converted into thermal energy, raising the temperature of the supply air stream. Bypassing this thermodynamic feedback causes the ideal-loads simulation to slightly underestimate space-cooling loads and overestimate space-heating requirements. Additionally, variable-flow pump power estimation suffers from the simplified assumption of a static hydronic loop temperature difference, which does not reflect real-world variable-flow dynamics controlled by differential pressure resets.
Third, commercial refrigeration is poorly suited for deterministic post-processing in food sales and food service archetypes, though it remains highly feasible for offices, schools, and hospitals where refrigeration is a minor plug load. In supermarkets and grocery stores, refrigeration display cases operate as powerful dehumidifiers and air coolers within the conditioned space. By drawing latent and sensible heat directly from the surrounding air, these open refrigerated cases significantly lower the active cooling load of the zone while simultaneously inflating the space-heating demand. Known as the refrigeration "case-credit" effect, this dynamic represents a major thermodynamic coupling. If refrigeration is estimated as a completely separate post-processing layer with no feedback into the thermal zone, the ideal-loads simulation will significantly overpredict cooling energy and underpredict heating energy, leading to systematic errors that can exceed 25% in the building's overall thermal energy balance.

Operational Gaps and Model Sensitivity Analysis

The reliance on deterministic, post-processed approximations introduces several structural uncertainties and empirical gaps that must be addressed to improve the reliability of urban-scale energy modeling.
First, a major source of error stems from the divergence in occupant water draw patterns. Standard code baselines assume highly regimented, uniform daily hot water draws. However, high-resolution field monitoring reveals massive stochastic variation in both daily draw volumes and water-use profiles, even among identical building typologies. Because peak water heating demands are driven by short, clustered draws rather than average daily totals, deterministic schedules struggle to accurately predict peak load timing on the grid. This limitation represents a critical gap for grid-interactive building integration studies.
Second, the structural thermal coupling of refrigeration equipment in food retail buildings remains an unaddressed limitation. In typical supermarkets, open display cases transfer substantial sensible and latent cooling loads directly into the zone air volume. Without a coupled thermal feedback loop, the simplified UBEM will model a highly distorted interior environment. To resolve this, future reduced-order algorithms must incorporate simplified, non-geometric coupling coefficients. These coefficients should inject a sensible cooling gain into the zone based on a fraction of the post-processed refrigeration power, establishing a pseudo-coupling without requiring full equipment modeling.

Third, the assumption of static temperature differences ($\Delta T_{\text{air}} = 11.1 \text{ K}$ and $\Delta T_{\text{water}} = 5.5 \text{ K}$) overlooks the performance impacts of modern dynamic HVAC controls. Real-world systems continuously reset supply air and chilled water temperatures to minimize fan and compressor energy under partial loads. Assuming a static temperature difference forces the SFP and SPP formulations to calculate volumetric flow rates that do not match the physical behavior of smart building systems. This limitation can be addressed by applying dynamic temperature-reset curves as a pre-processing layer to adjust the ΔT values based on outdoor air conditions before calculating the flow rates.
Fourth, historical empirical baselines introduce temporal misalignment risks when modeling modern urban sectors. The EUI end-use splits derived from CBECS 2018 reflect building operations prior to the widespread shift to hybrid work models. Consequently, the ratios of base-load energy to occupant-driven peak energy in commercial offices have changed. Applying unadjusted historical EUI splits to post-simulate current or future urban environments will overpredict occupant-driven auxiliary loads. Modellers must therefore apply scaling factors to adjust these baseline fractions, aligning the post-processed loads with modern, lower-occupancy profiles.

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